Of course, you can store standard file formats such as markdown, delimiter separated, JSON, or YML for a whole reproducibility approach. Note that we do not recommend keeping data under repos unless it's a data sample for testing.

Within a repository, how a team defines the expected folder structure for each project is generally less important than the consistent use of that structure. However, defining your project structure is still important. We recommend reading through *Big Book of MLOps* (Joseph Bradley, Rafi Kurlansik, Matthew Thomson, and Niall Turbitt, 2023, *Big Book of MLOps, second edition*, `https://www.databricks.com/resources/ebook/the-big-book-of-mlops`) to determine the best structure for your team or organization. As we will see in future chapters, MLflow and repositories are essential for **reproducible research**. In the world of data science and machine learning specifically, we want to ensure the reproduction of models and experiments.

Discussing data preparation

Generally, the first step for any data science project is to explore and prepare the data. We will refer to this process as moving the data from "Bronze" to "Silver" layers in reference to the Medallion architecture methodology. You might think of this type of data transformation exclusively as a data engineering task, but it's also essential for data science and machine learning.

If you aren't familiar with this architecture terminology, the **Medallion architecture** is a data design pattern used to organize data logically in a warehouse. This architecture is also commonly called "multi-hop" architecture. It aims to incrementally and progressively improve the structure and quality of data as it flows through each layer. The Medallion architecture has three layers: Bronze, Silver, and Gold, listed as follows:

- The **Bronze** layer is the raw data layer. It contains all the raw, unprocessed data ingested from the source systems. This data still needs to be cleaned or transformed.
- The **Silver** layer is the validated data layer. It contains data that has been cleaned and is subject to various validation and transformation steps. This data is ready to be used for analysis and modeling.
- The **Gold** layer is the enriched data layer. It is the highest level and contains data enriched with additional information, such as business intelligence metrics and key performance indicators, to meet the requirements of the business users.

The Medallion architecture is a flexible and customizable architecture that can meet the specific needs of each organization. The Medallion architecture is compatible with the concept of Data Mesh. Data Mesh is an architectural and organizational paradigm to ensure value from data. Lakehouse and Data Mesh are complementary, paradigms. See *Further reading* for blog posts on leveraging a data mesh with the DI Platform. This distributed data architecture enables organizations to unlock the value of their data by making it accessible and usable by everyone in the organization.